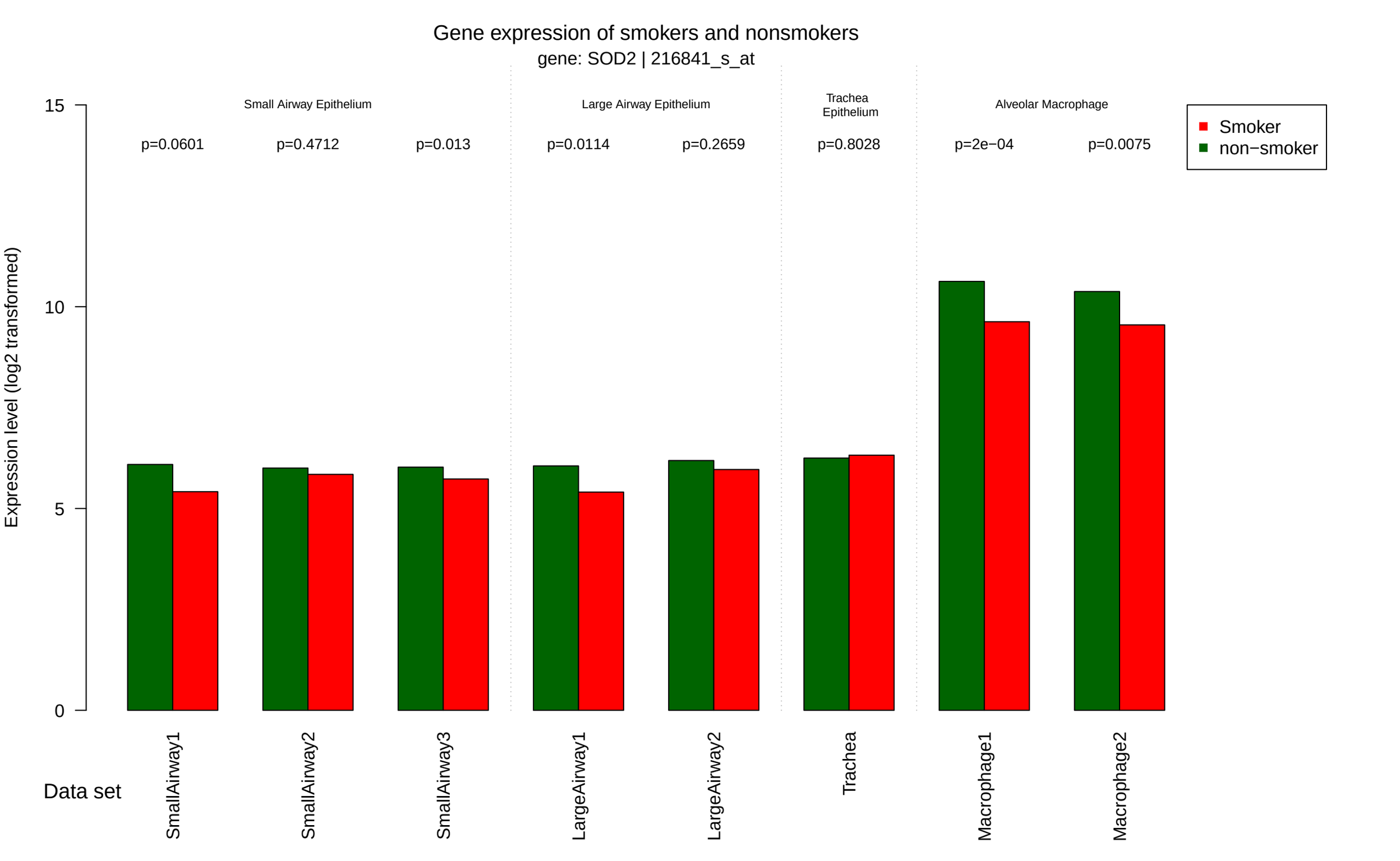
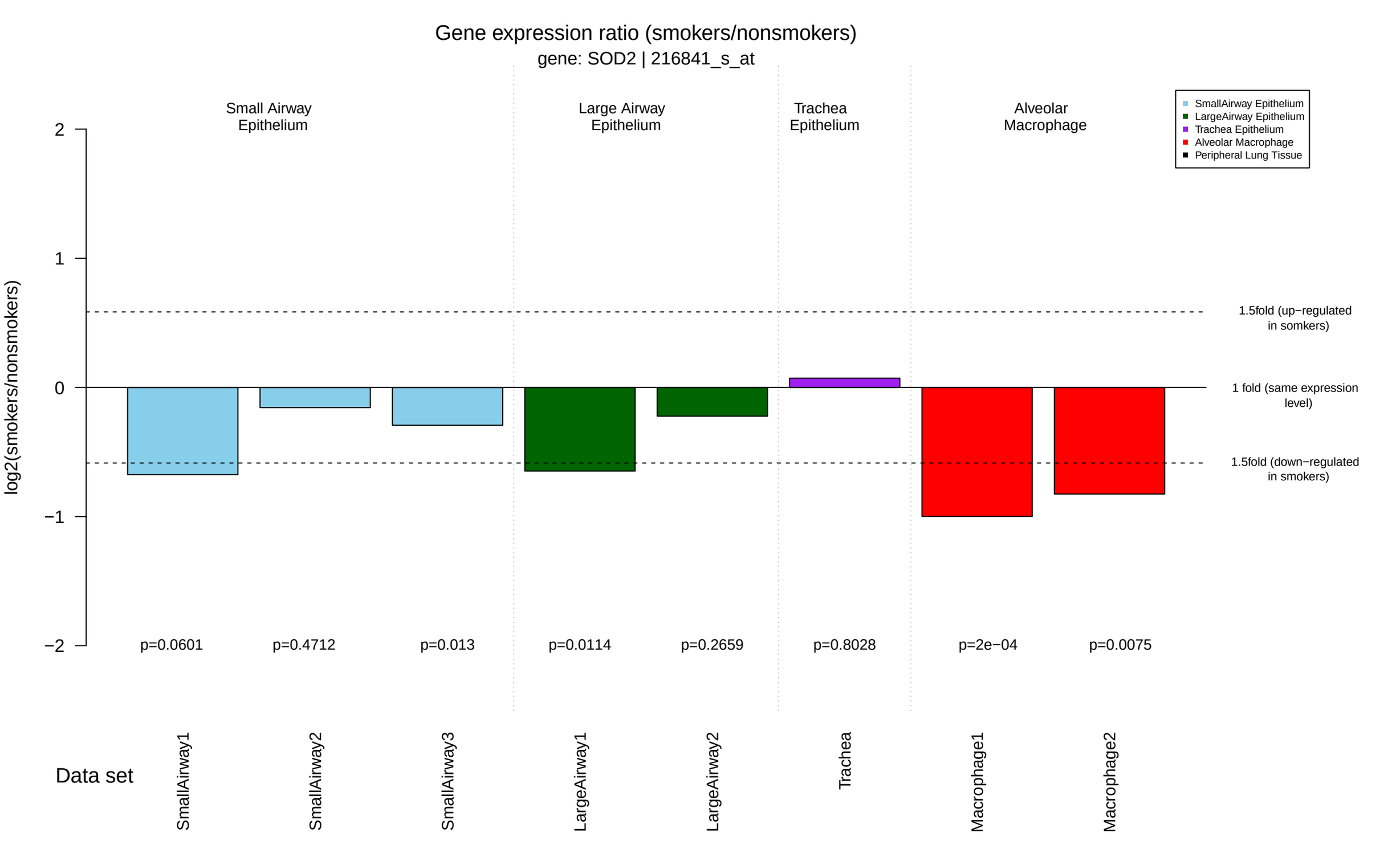
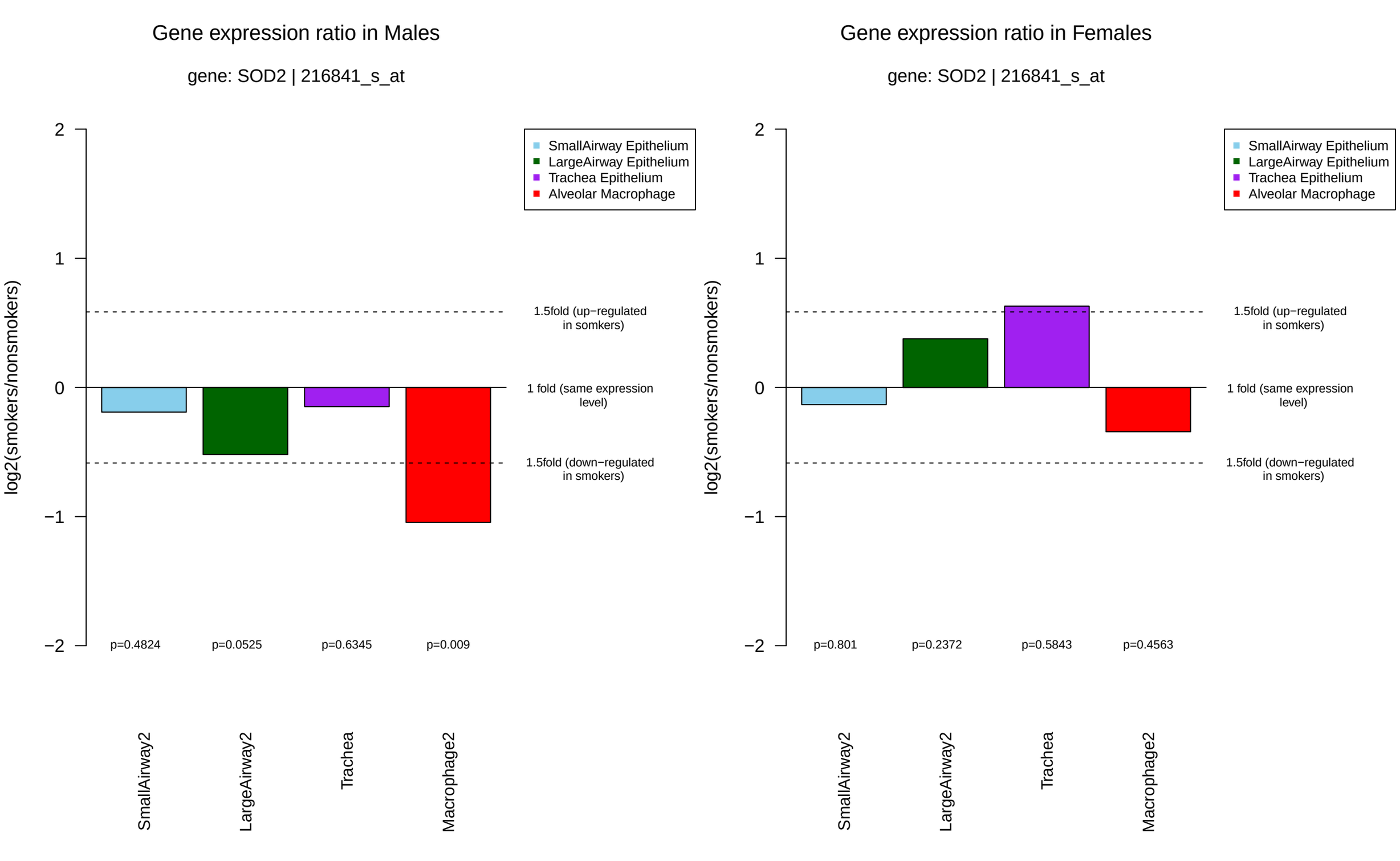
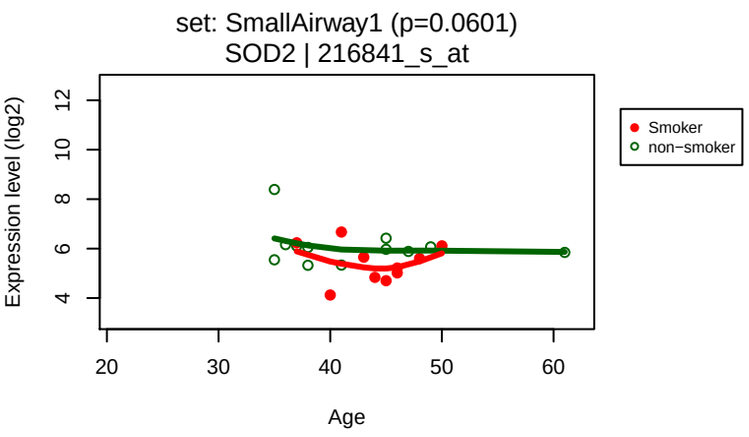
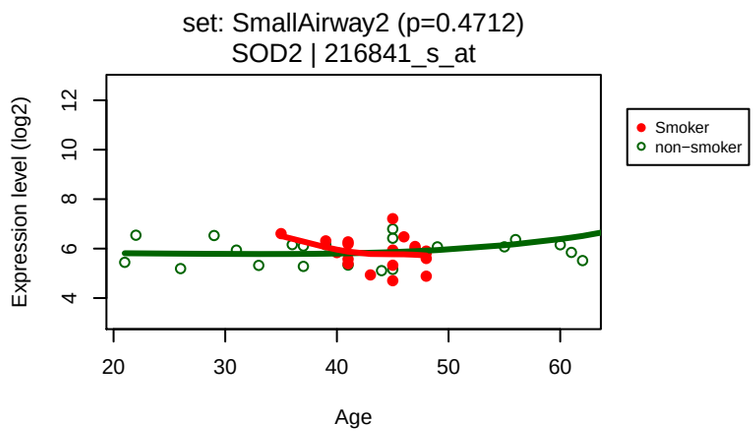
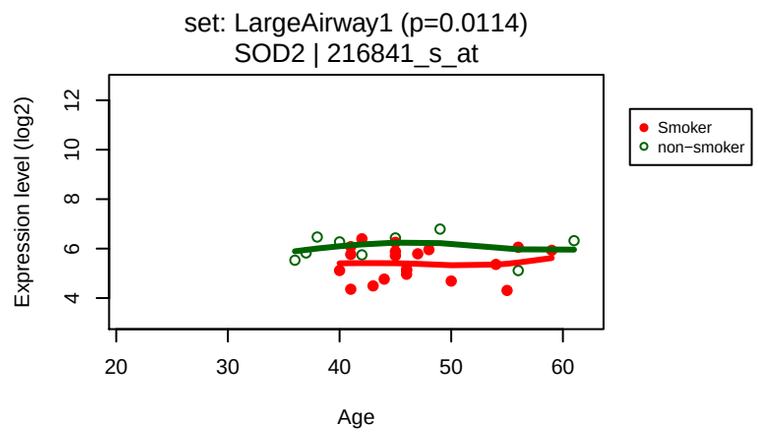

Gene expression of smokers and nonsmokers
gene: SOD2 | 216841_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0601
p=0.4712
p=0.013
p=0.0114
p=0.2659
p=0.8028
p=2e−04
p=0.0075
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

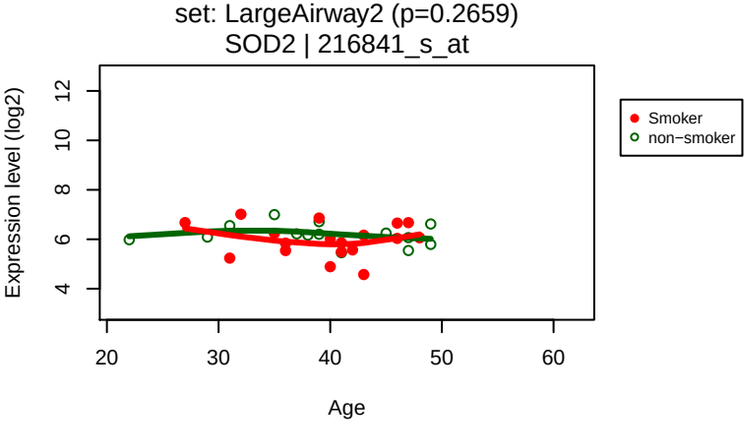

Gene expression ratio (smokers/nonsmokers)
gene: SOD2 | 216841_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0601
p=0.4712
p=0.013
p=0.0114
p=0.2659
p=0.8028
p=2e−04
p=0.0075
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

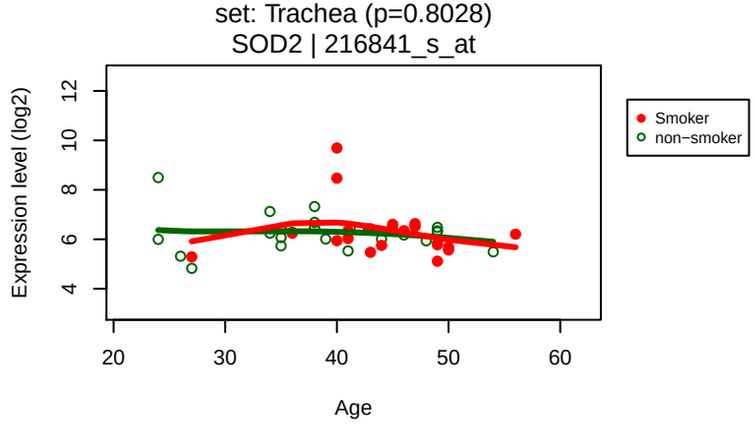

Gene expression ratio in Males
gene: SOD2 | 216841_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4824
p=0.0525
p=0.6345
p=0.009
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SOD2 | 216841_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.801
p=0.2372
p=0.5843
p=0.4563
SmallAirway2
LargeAirway2
Trachea
Macrophage2

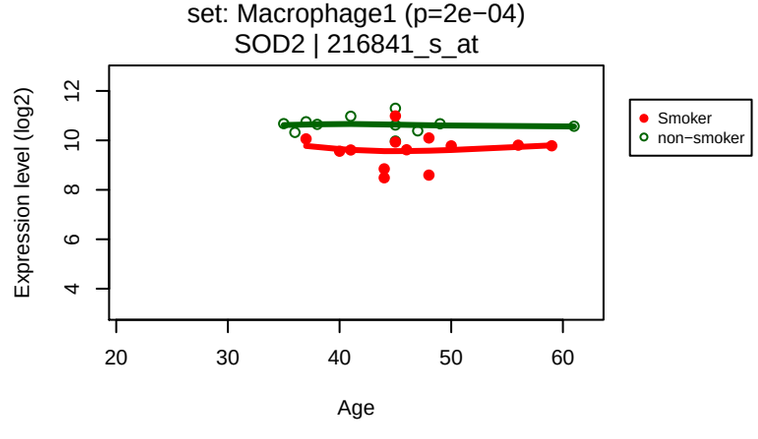

set: SmallAirway1 (p=0.0601)
SOD2 | 216841_s_at
Expression level (log2)
Age
Smoker
non−smoker

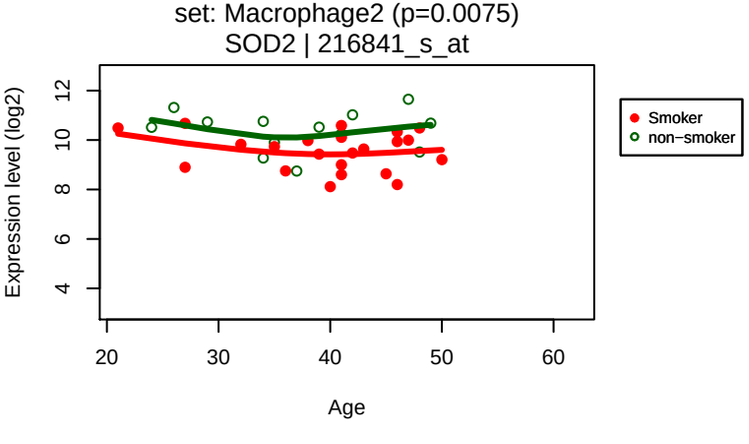

set: SmallAirway2 (p=0.4712)
SOD2 | 216841_s_at
Expression level (log2)
Age
Smoker
non−smoker

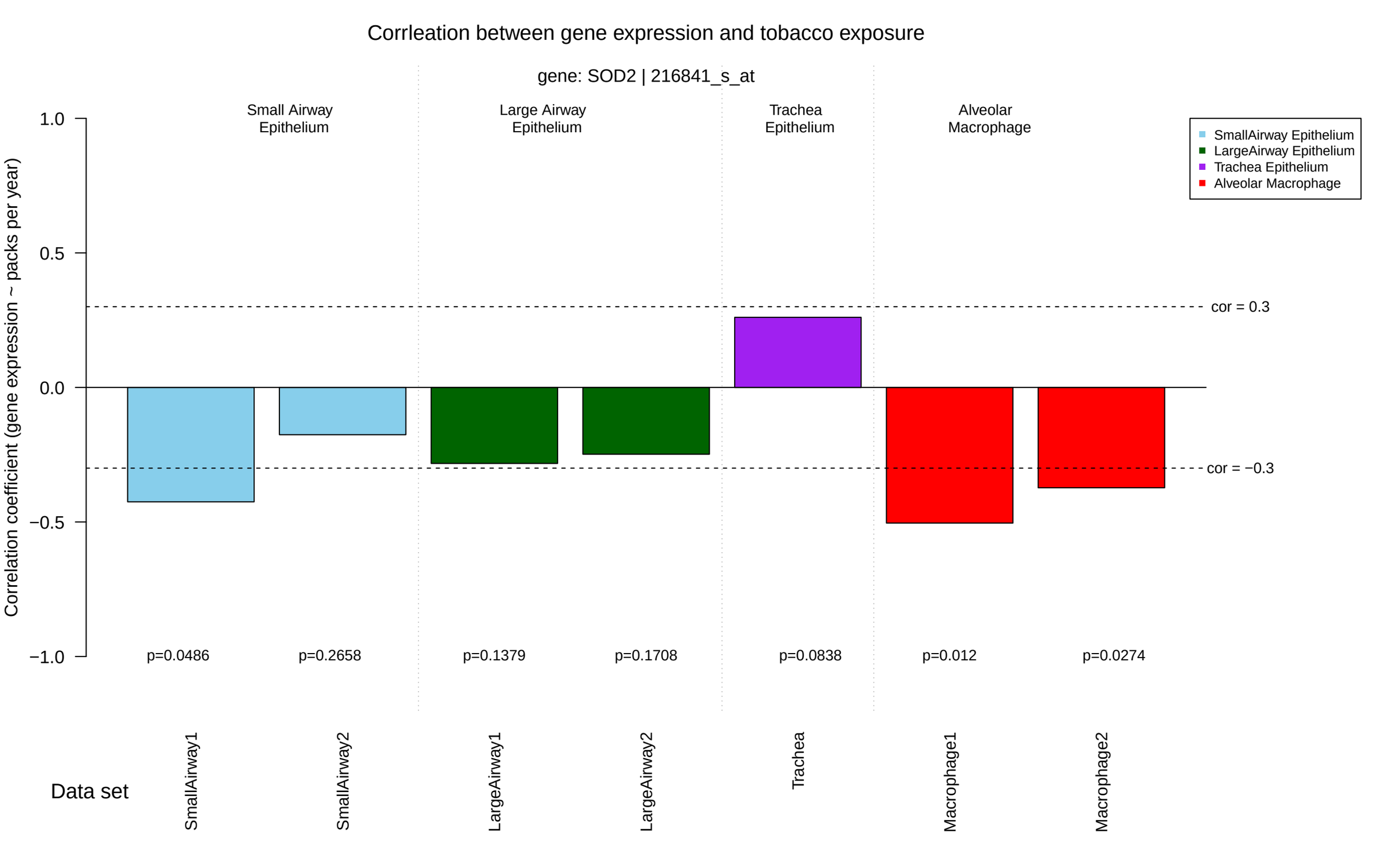

set: LargeAirway1 (p=0.0114)
SOD2 | 216841_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2659)
SOD2 | 216841_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8028)
SOD2 | 216841_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=2e−04)
SOD2 | 216841_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0075)
SOD2 | 216841_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SOD2 | 216841_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0486
p=0.2658
p=0.1379
p=0.1708
p=0.0838
p=0.012
p=0.0274
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2